Provenance Ledger record certificate

Issued 2026-10-17T23:01:23.851Z by Provenance Ledger

This record shows how this document was written in this environment; it does not detect AI, certify originality, or prove what the author consulted elsewhere.

The record

Record id	`9196354d75f45991c3792bd3c17e4e61b90cc91f7c2d4c2957c08e4fbd184135` (SHA-256 of the record's first event)
Tenant	`tp-06e272def2eb0da0477415b7c1c7e140`
Events	28, in 3 capture session(s), contract ledger-event/v0.1
Chain head	`event 28, 0f332608296cab8c5fadd68ed4fee44e2a81d2f16f45cdd76767f2ddbed5e5ea`
Head received	2026-10-17T19:22:52.143Z

Where the head is anchored

Sealed hour	2026-10-17T19:00:00.000Z to 2026-10-17T20:00:00.000Z, 1 events of the tenant
Hour root	`473f1754a75faef5cb776de4cbf6af0eccb8bb3661819ca24df448f7387a7ae9`
Root record SHA-256	`9e01c95a4806ea7a98e84eb33889d4413d508dded00614a617cd7cfa0bd0f94f`
Anchored root	`5bb1240ef6b878bf3d8ba32e60166bf5c161c145d070e05b73910576c440901c`

Authority	Time
RFC 3161, DigiCert	2026-10-17T20:07:23.000Z
RFC 3161, Sectigo	2026-10-17T20:07:23.000Z
OpenTimestamps, Bitcoin	block 971906, mined 2026-10-17T20:45:01.000Z

Bitcoin block	`971906, hash 00000000000000000000815765314ea275621058cf1885debd0b11ca7c7764b8`
Merkle root	`9fba68960c223bf8da74faed27b013d84803073329604ead7c24528d6ef6b6ab` (header byte order)

The certificate

Certificate id	`urn:uuid:ed65828a-9107-416d-924d-1b907e29702b`
Contract	record-certificate/v0
Issuer key	`Ed25519 d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a`
Issuer key id	`ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e`
Certificate SHA-256	`0e637203df645737f0df7bd115224b2e90ccb6532e34aad91bb58cf83fd6a2ba` (of certificate.json, exactly as attached)

How to check it

This PDF carries three attached files: certificate.json, the signed record, byte for byte; certificate.sig, the issuer's 64-byte Ed25519 signature of those bytes; and verify.sh, the commands below. Run them in a directory holding the two files, with the authorities' root certificates from anchors/tsa/ of the anchors record for step 5. Each check prints ok or FAIL; nothing in them needs this ledger. The same commands are in certificate.json under procedure. Last, compare the block hash with any Bitcoin node or explorer. The certificate's SHA-256 is listed in the public submission index (submission-index/v0).

```
# Record certificate check: bash with sed, sha256sum, base64, grep, cut, fold, tac and OpenSSL 3.
# In a directory with certificate.json and certificate.sig, and for step 5 the authorities' root
# certificates from anchors/tsa/ of the anchors record (https://github.com/Korentsvit/fundingmachine-anchors).
set -euo pipefail
unhex() { printf "$(tr -d ' \n' | sed 's/../\\x&/g')"; }
step() { printf '%s%s%s' "$1" "$x" "$2" | unhex | sha256sum | cut -c1-64; }
check() { if [ "$1" = "$2" ]; then echo "ok    $3"; else echo "FAIL  $3"; exit 1; fi; }
```

```
field() { grep -o "\"$1\":\"[^\"]*\"" "$2" | sed -n "${3:-1}p" | cut -d'"' -f4; }
echo "certificate SHA-256 $(sha256sum certificate.json | cut -c1-64)"
# 1. The issuer's key signed exactly these bytes.
printf '%s' 302a300506032b6570032100d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a | unhex > issuer.der
openssl pkeyutl -verify -pubin -keyform DER -inkey issuer.der -rawin -in certificate.json -sigfile certificate.sig
check "$(printf '%s' d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a | unhex | sha256sum | cut -c1-64)"
ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e '1. the issuer key id'
# 2. The record's head event, 0f332608296c..., is in the root of the hour 2026-10-17T19:00:00.000Z.
x=0f332608296cab8c5fadd68ed4fee44e2a81d2f16f45cdd76767f2ddbed5e5ea
x=$(step '00' '')
check "$x" 473f1754a75faef5cb776de4cbf6af0eccb8bb3661819ca24df448f7387a7ae9 '2. the head event is in the hour root'
# 3. The hour's root record carries that root and is signed by the issuer's key.
field root_record_b64 certificate.json | base64 -d > root-record.json
check "$(sha256sum root-record.json | cut -c1-64)" 9e01c95a4806ea7a98e84eb33889d4413d508dded00614a617cd7cfa0bd0f94f '3. the hour root
record'
check "$(field root root-record.json)" 473f1754a75faef5cb776de4cbf6af0eccb8bb3661819ca24df448f7387a7ae9 '3. the record carries the
hour root'
check "$(field key_id root-record.json)" ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e '3. the record names the
issuer key'
sed 's/"sig":"[^"]*",//' root-record.json > root-record-unsigned.json
printf 'ledger-root/v0:%s' "$(sha256sum root-record-unsigned.json | cut -c1-64)" > root-message
printf '%s==' "$(field sig root-record.json)" | tr '_-' '/+' | base64 -d > root-record.sig
openssl pkeyutl -verify -pubin -keyform DER -inkey issuer.der -rawin -in root-message -sigfile root-record.sig
# 4. The root record is in the hour's anchored root.
x=9e01c95a4806ea7a98e84eb33889d4413d508dded00614a617cd7cfa0bd0f94f
x=$(step '00' '')
check "$x" 5bb1240ef6b878bf3d8ba32e60166bf5c161c145d070e05b73910576c440901c '4. the root record is in the anchored root'
# 5. Each RFC 3161 token stamps the anchored root; checked against its authority's root certificate at its time.
field token_b64 certificate.json 1 | base64 -d > digicert.tsr
if [ ! -f digicert.pem ]; then echo 'skip  5. DigiCert: digicert.pem is not here'; elif openssl ts -verify -digest
5bb1240ef6b878bf3d8ba32e60166bf5c161c145d070e05b73910576c440901c -in digicert.tsr -CAfile digicert.pem -attime 1792267643 >/dev/null
2>&1; then echo 'ok    5. DigiCert stamped the anchored root at 2026-10-17T20:07:23.000Z'; else echo 'FAIL  5. DigiCert stamped the
anchored root at 2026-10-17T20:07:23.000Z'; exit 1; fi
field token_b64 certificate.json 2 | base64 -d > sectigo.tsr
if [ ! -f sectigo.pem ]; then echo 'skip  5. Sectigo: sectigo.pem is not here'; elif openssl ts -verify -digest
5bb1240ef6b878bf3d8ba32e60166bf5c161c145d070e05b73910576c440901c -in sectigo.tsr -CAfile sectigo.pem -attime 1792267643 >/dev/null
2>&1; then echo 'ok    5. Sectigo stamped the anchored root at 2026-10-17T20:07:23.000Z'; else echo 'FAIL  5. Sectigo stamped the
anchored root at 2026-10-17T20:07:23.000Z'; exit 1; fi
# 6. The anchored root reaches the merkle root of Bitcoin block 971906, whose header is below.
x=5bb1240ef6b878bf3d8ba32e60166bf5c161c145d070e05b73910576c440901c
x=$(step '' 'ed94c20d219f04fd89648e253aad73f1')
x=$(step '' '5dae29b983101071')
x=$(step '' '1a8c7ee1cb806d594c04db3b301b96e3')
x=$(step 'c766cda6886c31d2849b2755223380a23ec8e046a3cf6dd435c852766c0f4cde' '')
x=$(step '' 'ffc8ba1a8f2f9f8bb5aa000942beaf70ceb9887b67d50850a4afe6484c98d76d')
x=$(step '' 'a0b16f712f03395019f0751a1cf6df69cd1596f8d65534b39e1574834950b84d')
x=$(step '6ad3d57b' '8a41e2ddb0f43cad')
x=$(step '' 'f525c364c3f7d120c409370105e4ed26ee4a30bd0df9f44c826321875c114916')
x=$(step '' '288c320010fa992ff1555ea107fb6616cc4c951996b85bf4ee58b0cad947f537')
x=$(step '' '8672e8ef36f062e080ed3e6236bd55d56e1f85fa5107b228473adf8e86a83765')
x=$(step 'b680c0126d6283a0a8173da7a3c4d43cae451ca3da9c014bd5358819fd62602d' '')
x=$(step '' '210dbe7090c32eea314f7f7de3040938e1b008b542370b42a7ab797ca63239d1')
x=$(step '7ce225521f55b05f07dfea396c013511230f4c44353e7839c984d7ea234282d2' '')
x=$(step 'e4dec1e6b05b632560d09383ddd4badc03b020d0f7175703c2c1fba0b2af429c' '')
x=$(step '' 'cd2a7102238a51c27c1ead83b3dcb00b7fb071beb19f07c57b6854f2e5e8234c')
x=$(step '6d0962e1b8e3d1b7fa5126815dd89bceac9bbfeab5ec3005cebe444a98e4e40d' '')
x=$(step 'd9bb0a6149be8f8283e9ac6e503534f165dceb03ce06e58ab6a70cdf985e5d90' '')
x=$(step '1f6170075ed9b9f7d05681859b766cc488527e4ec5c5f2e4b31850e96cad3908' '')
x=$(step '7849fc080b85efc55f5b4d733c37e3569bed0c3ee00e2adcbca26f42d793b40f' '')
x=$(step '87285ec1031cc2221704ba7d79b07ecd1e20a8481b12f027c8fdaabbfdf9cbd6' '')
x=$(step '' 'ff9b0778cdaaaa72aa2fb897ed8e3c68f673d755a72cb6d1d5d2dc51ceb9d1ef')
x=$(step
'01000000018e9d076417d57e12f8b2d723da0cf078ab82bca56cb7f3b893fa16ad67cb190f0000000000feffffff028049010000000000160014afbdd4fff24d4028
2729546aeab42c1c053b89d8000000000000000000226a20' '81d40e00')
x=$(step '' '')
x=$(step '3b17773c19d50a735fcdb069eaaeb2a83f68ba20526bd08a4bed0230ee75367c' '')
x=$(step '' '')
x=$(step '00279762442ca1aeec0740cbbf5ba5ef556019c28c9e20b464a3dec4de21ed58' '')
x=$(step '' '')
x=$(step '' 'cd32dfcf470f04df470051a0281123b49a19644762aed3f3e957fb6972aa9633')
x=$(step '' '')
x=$(step '' '7cd461e502d6a8fd8d0fab9b70a57e42b64bdeb5ea19ac0f1e410be3db0ed02a')
x=$(step '' '')
x=$(step '6c56848ccb24cfd2e0347847d30e8488b7e04c8ae1fc80a3b09805fba62625db' '')
x=$(step '' '')
x=$(step 'c83e0528711cd0b82f805ff6ba9b8dfbfff3bfc1674687813a20dc34e882c513' '')
x=$(step '' '')
x=$(step '6e8c316bcbbc4d71ada286bcba53060a84e5903248043df6540c8258257dd399' '')
x=$(step '' '')
x=$(step '39780991aa628ce35b1f366232cb79e59159ddfa34cd5e4e3b3e157704a2988d' '')
x=$(step '' '')
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x=$(step '6210a7dcffafa159108480d6ea8c7759e2ac5c7745c0931abf8d84aa320fb50b' '')
x=$(step '' '')
x=$(step 'a6027692422ffc71d2b0888558cfe5fe694733c212494965fd3d9b55f943f6c8' '')
x=$(step '' '')
x=$(step '' 'd91325bc331a832672ee8607e5e16e4f7a9403f4bb8d18baf03a0106ec88b51e')
x=$(step '' '')
x=$(step '' '9bf82838d54a89c9ced63ead597a5d450805f811da17384b095748bba219fc73')
x=$(step '' '')
x=$(step '' '3e0e09463e0faefd3a555fd09396c5c75a0cf4c88287eab5628e8f6c1bc2ebb4')
x=$(step '' '')
check "$x" 9fba68960c223bf8da74faed27b013d84803073329604ead7c24528d6ef6b6ab '6. the anchored root reaches the merkle root of block
971906'
h=00e03c22e221f570f6710e93ab9f53a433f5bc31f7d6dd15adf5000000000000000000009fba68960c223bf8da74faed27b013d84803073329604ead7c24528d6ef
6b6ab4dded36a93000217d0f1de23
check "${h:72:64}" 9fba68960c223bf8da74faed27b013d84803073329604ead7c24528d6ef6b6ab '6. the block header carries that merkle root'
check "$(printf '%s' "$h" | unhex | sha256sum | cut -c1-64 | unhex | sha256sum | cut -c1-64 | fold -w2 | tac | tr -d '\n')"
000000000000000000000815765314ea275621058cf1885debd0b11ca7c7764b8 '6. the header is block
000000000000000000000815765314ea275621058cf1885debd0b11ca7c7764b8'
echo "Last, compare block 971906 with any Bitcoin node or explorer: its hash must be
000000000000000000000815765314ea275621058cf1885debd0b11ca7c7764b8."
```